

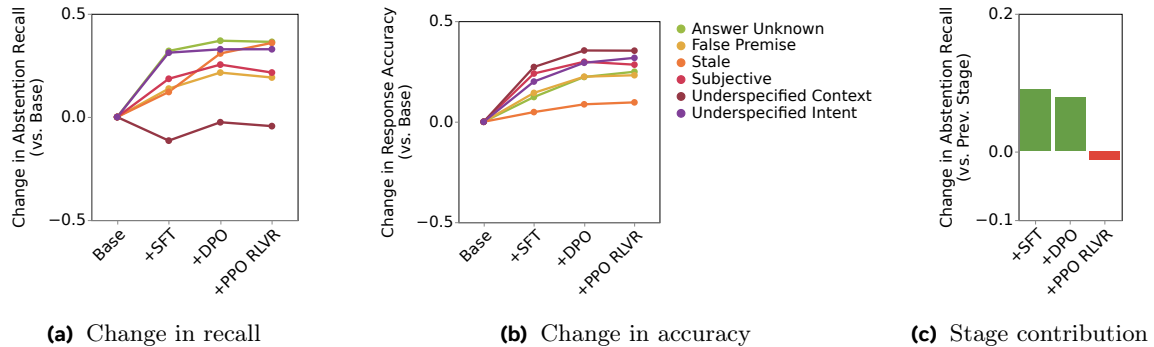


Figure 5 Post-training improves response accuracy and abstention recall, but not for underspecified context. (a) Change in abstention recall of Tülu checkpoints vs. Llama 3.1 Base 8B. (b) Change in response accuracy. See Appendix D for precision and F1 score. (c) Contribution of each post-training stage to change in recall: RLVR degrades abstention.

Improved accuracy does not imply improved abstention. While model capabilities often increase in line with one another, in Fig. 4 we see that improving response accuracy does not necessarily imply improved abstention performance. On the GSM8K-Abstain dataset of underspecified math problems, we see a positive correlation, while faced with FreshQA questions that are unanswerable given models’ pre-training cutoff, improving correctness correlates with degraded abstention.

Underspecification, subjectivity, and false assumptions are key challenges. In Fig. 3c we show the performance distribution aggregated by different scenarios for the overall best model, Qwen 2.5 32B. Questions that do not provide sufficient evidence, contain incorrect assumptions, or have no universally agreed upon answer are persistently challenging and induce variable abstention performance. Instead of clarifying, expressing uncertainty, or pointing out incorrect assumptions, models inappropriately respond definitively (see Appendix F for qualitative examples). This suggest models may not be appropriately reasoning about evidence and claims in these scenarios.

4.2 Post-training instills select abstention capabilities

The key capabilities of contemporary LLMs are instilled during post-training fine-tuning. In a typical pipeline, a base language model (i.e., a next token predictor) might undergo supervised fine-tuning (SFT) [80] to induce instruction following behavior, followed by optimization on human feedback [81, 80, 82] to improve user satisfaction, and more recently optimization with a verifiable reward signal to improve correctness on reasoning-focused tasks [77]. While each of these stages are critical components of LLM performance, it is unclear how they contribute to abstention.

To understand this, we test the Tülu 3 [77] series of model checkpoints released at various stages in the post-training lifecycle. Relative to a Llama 3.1 base model, we evaluate the change in abstention performance induced by each successively applied post-training stage: SFT, followed by direct preference optimization on preference data (DPO) [82], and finally proximal policy optimization with a verifiable reward (PPO RLVR) [81, 80, 77].

Post-training provides limited improvement for abstaining given underspecified contexts. Overall, we observe that abstention recall (Fig. 5a) and non-abstention response accuracy (Fig. 5b) tend to improve throughout SFT and DPO on most scenarios. A notable exception is underspecified context samples, on which we have previously seen that many models exhibit highly variable and often poor performance (see Section 4.1). Tülu post-training worsens abstention recall on underspecified contexts, with a sharp drop during SFT (Fig. 5a). Based on the composition of open post-training datasets [77], this may due to a general lack of underspecified context prompts. Increasing the representation of underspecified context prompts during SFT post-training may be a promising future direction for improving abstention performance.

Verifiable reward post-training degrades abstention. Comparing the relative change in abstention recall *between* each successive stage (Fig. 5c), we observe a surprising degradation in abstention after RLVR. The Tülu RLVR checkpoint has undergone reinforcement learning-based fine-tuning on math and verifiable instruction-following

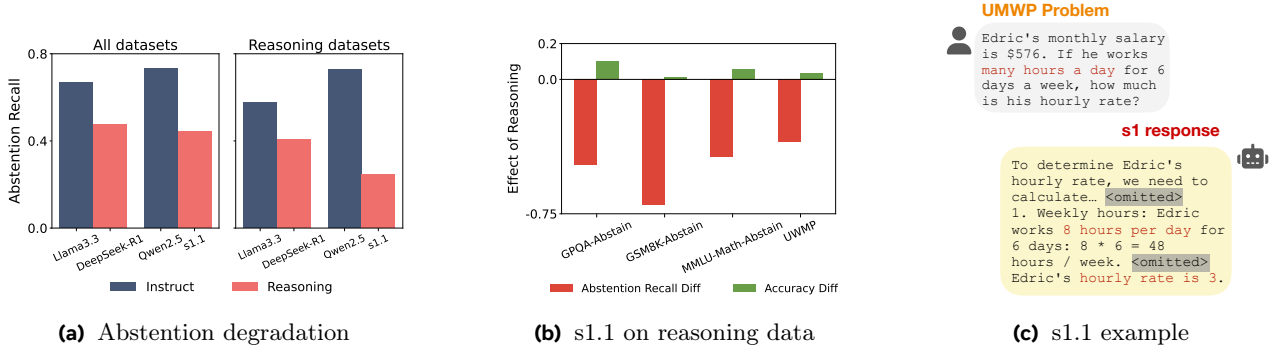


Figure 6 Reasoning models answer definitively when they shouldn't. (a) Comparing each reasoning LLM vs. its underlying instruct model (i.e. DeepSeek R1 Llama 70B Distill vs. Llama 3.3 70B; S1.1 32B vs. Qwen 2.5 32B), reasoning models exhibit worse abstention on all datasets (left), including reasoning datasets (right; GSM8K-Abstain, GPQA-Abstain, MMLU-Math-Abstain, UMWP). (b) For s1, reasoning boosts accuracy (green) while degrading abstention (red). (c) Example of s1 failure to abstain on a problem from UMWP by hallucinating missing context.

datasets in order to improve response correctness on reasoning tasks. We hypothesize that optimizing for the clear-cut verifiable reward signal has an undue influence on handling uncertain or unanswerable questions, motivating our study of reasoning interventions in Section 4.3.

We present additional evidence on the role of post-training in Appendix D, including consistent results both at 70B scale, and when comparing Llama 3.1 70B Base vs. Instruct, with the Instruct checkpoint incorporating repeated successive rounds of SFT and DPO [73].

4.3 Reasoning degrades abstention

Reasoning fine-tuning LLMs has improved their capabilities, especially in math, coding, and science. However, it is unclear whether these advances generalize to reasoning about evidence and uncertainty and identifying unanswerable questions. Our previous result (see Section 4.2) suggests the RLVR reasoning stage degrades abstention. Here, we use recent state-of-the-art reasoning LLMs DeepSeek R1 Distill (Llama 70B) and s1.1 (32B) to systematically study this question. We compare DeepSeek R1 Distill and s1.1 to their underlying instruction-tuned models (Llama 3.3 70B and Qwen 2.5 32B, respectively), allowing us to isolate the effect of reasoning from confounders such as model architecture or pretraining data. We use tokenizer templates which start generation with a “start-of-thinking” token, allocating 4k tokens for reasoning before forcing a final answer in an additional 4k tokens. We also additionally explore the effect of scaling test-time compute by varying the reasoning token budget. Unless otherwise stated, our evaluation only considers the final response (rather than the reasoning trace) when determining abstention. See Appendix C for further details.

Reasoning models struggle to abstain, even in math and science domains. Comparing reasoning fine-tuned models to their underlying instruction-tuned models, Fig. 6a (left) shows that reasoning models exhibit worse abstention performance than their non-reasoning counterparts, as measured by recall, with F1 score showing the same trend (see Appendix D).

Next, we focus on AbstentionBench math and science datasets where reasoning models have been shown to excel. On our underspecified variants of popular reasoning datasets and on UMWP (see Section 3.1), Fig. 6a (right) shows that reasoning models exhibit degraded abstention, even on these domains on which models were explicitly trained.

Despite boosting accuracy, reasoning degrades abstention. The degraded abstention performance detailed above isn't the result of unsuccessful reasoning fine-tuning. Indeed, in Fig. 6b we observe that s1—as expected—exhibits improved response accuracy over Qwen 2.5 for reasoning benchmarks, but this comes at the cost of significantly decreased abstention performance. We show analogous results for DeepSeek R1 Distill in Appendix D. Qualitatively, we find that models often hallucinate the missing problem context, as we show in Fig. 6c and Appendix F.

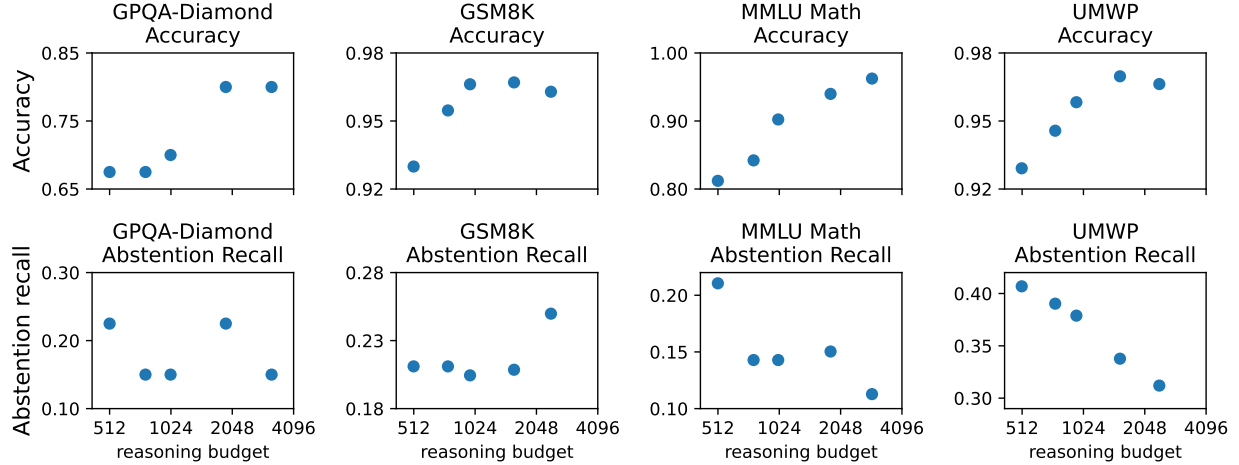


Figure 7 Increasing reasoning budget in s1.1 improves accuracy and hurts abstention. We evaluate test-time scaling of s1.1 by interrupting its thinking chain after 512, 768, 1024, 2048, and 4096 tokens. **Top row:** response accuracy. **Bottom row:** abstention recall.

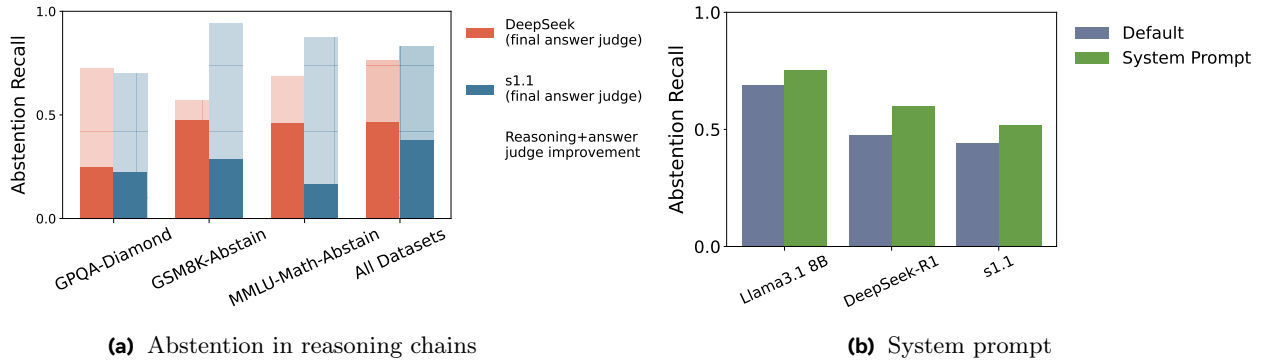


Figure 8 (a) While reasoning chains contain expressions of uncertainty, reasoning models still provide definitive answers. **(b)** A system prompt describing scenarios where models should not respond boosts abstention for both standard and reasoning LLMs.

A key factor in supporting reasoning performance is test-time compute budget [14], i.e., the number of tokens dedicated to reasoning prior to final answer generation. To isolate the effect of test-time scaling on abstention, we follow Muennighoff et al. [14] in evaluating the reasoning models with a varying maximum reasoning budget (from 512 to 4096 tokens), after which we force the generation of the “start-of-final-answer” token. Fig. 7 shows model accuracy (top row) and abstention recall (bottom row) for s1 against average empirical reasoning budget on four reasoning datasets. As reasoning budget increases, accuracy improves with abstention either not improving (GSM8k-Abstain) or worsening (UMWP). We observe the same trend for DeepSeek R1 Distill, but mixed results when experimenting with o1’s “reasoning effort” hyperparameter (see Appendix D)—though we unfortunately lack transparency into the mechanism behind this closed API. We hypothesize that the negative consequences of increasing time-time compute result from reward model misspecification, where models are biased to provide definitive and confident responses.

Models express uncertainty in reasoning chains, but still provide a definitive final answer. By default we only evaluate reasoning models’ final answer, though we experiment with additionally passing the lengthy reasoning traces to the LLM judge. In Fig. 8a, we see that reasoning traces do contain increased expressions of uncertainty, but despite this, models continue to provide a definitive final response. Incorporating the reasoning chain also degrades abstention precision (see Appendix D), particularly for models such as s1 that are explicitly optimized to emit “Wait” tokens and extensive self-critique. While the uncertainty in reasoning traces is